Patents

Deodorizing filter

Landscapes

Exhaust Gas Treatment By Means Of Catalyst

Solid-Sorbent Or Filter-Aiding Compositions

Catalysts

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JP2981714B2
Japan

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Other languages: Japanese

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Current Assignee : Nichias Corp

Worldwide applications

1995 JP

Application JP7094268A events

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1996-10-15	Publication of JPH08266866A
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Info: Cited by (4), Legal events, Similar documents, Priority and Related Applications

External links: Espacenet, Global Dossier, Discuss

Claims (3)

Hide Dependent
translated from Japanese

(57) [Claims] 1. A deodorizing filter comprising a deodorizing catalyst is malodorous untreated air a long period of use it may become feel heterogeneous unpleasant odor is supported on the gas-reciprocal applications catalyst support, SiO_2 / Al A zeolite powder having a $_2O_3$ ratio of 15 or more is used in an amount of 15 to 40 per 100 parts by weight of the deodorizing catalyst.
A deodorizing filter characterized by being supported on a catalyst carrier together with the deodorizing catalyst in a ratio of parts by weight . 2. A catalyst support for a gas-phase reaction having a honeycomb structure made of inorganic fiber paper having a high porosity, a Mn-Cu-based deodorizing catalyst and a zeolite powder having a SiO_2 / Al_2O_3 ratio of 15 or more, and 100 wt. Zeolite powder per part 15
A deodorizing filter supported at a ratio of 4 0 40 parts by weight . 3. A catalyst carrier for a gas-phase reaction having a honeycomb structure made of inorganic fiber paper having a high porosity, and a Mn—Cu-based deodorizing catalyst and a zeolite powder having a SiO_2 / Al_2O_3 ratio of 15 or more being 100 parts by weight of a deodorizing catalyst. Per zeolite powder 15 ~
40 parts by weight , the zeolite powder is supported at a higher density on the core than the surface of the carrier, and the deodorizing catalyst is supported on the surface of the carrier at a higher density than the core. A deodorizing filter characterized by the following.

Description

translated from Japanese

DETAILED DESCRIPTION OF THE INVENTION

[0001]

BACKGROUND OF THE INVENTION 1. Field of the Invention The present invention relates to a deodorizing filter used for deodorizing air having malodor, that is, a deodorizing catalyst having an action of accelerating the decomposition of malodorous components in air is supported on a carrier for gas phase reaction. The present invention relates to a deodorizing air filter.

[0002]

2. Description of the Related Art In a house or other building, not only in a special place such as a toilet, but also when ventilation is insufficient, an unpleasant odor may be felt. The use of agents has been practiced from an early age.
However, these odor countermeasures are not only ineffective but also temporary. Therefore, a deodorizing filter in which a deodorizing catalyst having a function of accelerating the decomposition and deodorization of malodorous components is supported on a catalyst carrier for a gas phase reaction has attracted attention as a new air deodorizing means, and some air conditioners and air purifiers have already been used. It is mounted on refrigerators and the like, and is also practically used as an independent deodorizer.

[0003] A deodorizing filter that decomposes malodorous components in the air by utilizing the action of a deodorizing catalyst does not require labor and cost for maintenance, and a constant deodorizing effect is guaranteed for a long time. Much better than that. However, it has been pointed out that if this deodorizing filter is used for a long time, an unpleasant odor different from the unpleasant odor of the untreated air may be felt. This is presumably because some of the plurality of

decomposition products generated by decomposition of the malodorous component under the action of the deodorizing catalyst become new odor-causing substances. The reason why a new odor due to the use of a deodorizing filter is not felt at all at the beginning of use of the filter but is used for a long period of time, Not yet elucidated.

[0004]

SUMMARY OF THE INVENTION It is an object of the present invention to provide a conventional deodorizing filter having the above-mentioned drawbacks, which may cause discomfort to the user or replace it despite having sufficient catalytic activity. An object of the present invention is to provide a deodorizing filter that does not give off odor even when used for a long period of time and has a longer lifespan than conventional products in view of the necessity.

[0005]

SUMMARY OF THE INVENTION The deodorizing filter which the present invention has succeeded in is provided on a catalyst carrier for a gas-phase reaction together with a deodorizing catalyst having the above-mentioned drawbacks by using SiO_2 / Al_2O_3 ratio of 15 or more zeolite powder is made by supporting a proportion of 15 to 40 parts by weight deodorizing catalyst 100 parts by weight per zeolite.

Particularly preferably, the deodorizing filter of the present invention comprises a Mn-Cu-based deodorizing catalyst and a SiO_2 / Al catalyst on a catalyst support for a gas phase reaction having a honeycomb structure made of a high porosity inorganic fiber paper.
A zeolite powder having a $_2O_3$ ratio of 15 or more is supported at the above ratio .

The zeolite supported on the carrier together with the deodorizing catalyst in the ozone filter according to the present invention is SiO_2 / The Al_2O_3 ratio is 15 or more. This zeolite with a high SiO_2 ratio is more carbonized than water vapor, in contrast to the usual zeolite with a SiO_2 / Al_2O_3 ratio of about 2 to 15 which is strongly hydrophilic and used as a drying agent. It is known that hydrogen compounds are well adsorbed (hereinafter, this highly hydrophobic zeolite is referred to as hydrophobic zeolite). The composition of a typical example is as follows: $HalSi_{35}O_{72}$, $HalSi_{25}O_{52}$ and the like (see Japanese Unexamined Patent Publication No. Sho 60-501495 and Japanese Unexamined Patent Publication No. Hei 1-331457). Commercially available products that are easily available include those having a SiO_2 / Al_2O_3 ratio of about 50 to 500.

[0008] The deodorizing filter of the present invention in which a hydrophobic zeolite powder is supported on a carrier together with a deodorizing catalyst does not cause a new off-flavor even after long-term use. For unknown reasons, the adsorption of hydrophobic zeolites to atmospheric odor components or their decomposition products in the vicinity of the deodorizing catalyst helps to prevent the generation or release of secondary odor-causing substances. It is assumed that

[0009] responsible Jiryou of hydrophobic zeolite in the deodorizing filter of the present invention, the deodorizing catalyst 100 parts by weight per about 1 5 to 40 parts by weight. If the amount of the hydrophobic zeolite is not sufficient, the generation of offensive odor cannot be completely prevented. On the other hand, when the hydrophobic zeolite is supported more than necessary, the amount of the deodorizing catalyst to be supported is limited, and the deodorizing performance of the deodorizing catalyst is reduced.

Hereinafter, the deodorizing filter of the present invention will be described while explaining the production method. A deodorizing catalyst supported on a deodorizing filter generally converts malodorous substances into other substances having low odor intensity by promoting various reactions mainly oxidizing reactions by oxygen in the air, thereby deodorizing or reducing odors. As typical examples, Mn-Cu-based, Fe-Mn-Au-based, and Ti-Mn-based are known. The deodorizing catalyst used in the deodorizing filter of the present invention may be any of these, and is not limited. However, when using a deodorizing catalyst having no problem of emitting an off-odor after long-term use as described above, the present invention Needless to say, it does not make much sense to take countermeasures.

As the hydrophobic zeolite, those having the above-mentioned composition, preferably having a particle size of about 0.5 to 30 μm, are used. Zeolites coarser than those in the above particle size range are not preferred because they hardly enter the fine voids of the carrier.

The carrier for supporting the deodorizing catalyst and the hydrophobic zeolite is not particularly limited as long as it is suitable as a carrier for the deodorizing catalyst and can support the hydrophobic zeolite. Such a carrier having a honeycomb structure (for example, one described in JP-A-59-10345) is particularly suitable as a carrier in the deodorizing filter of the present invention.

[0013] The above-mentioned honeycomb structure carrier comprises alumina fiber,
It is made of paper made of inorganic fibers such as silica-alumina fiber, glass fiber and rock wool and having a high porosity of 70 to 95%. As a commercial product, "Hanicle" (product of Nichias Corporation) is available. is there. The porosity is a value obtained from the following equation. Porosity (%) = (1−apparent density / true density) × 10
0

A hydrophobic zeolite is supported on the above-mentioned carrier together with a deodorizing catalyst. The effect of co-supporting the hydrophobic zeolite appears even in a small amount.
It is desirable to carry it in the range of 40% by weight. Excessive amount of hydrophobic zeolite inhibits the action of the deodorizing catalyst.

The loading method is not particularly limited, but a typical method is a method in which both are loaded stepwise in the following order.

First, only the hydrophobic zeolite is supported. The hydrophobic zeolite to be supported is suspended in water together with an inorganic binder such as silica sol, alumina sol, titania sol and the like. At this time, it is desirable to use a minimum amount of the binder necessary for fixing the hydrophobic zeolite to the fiber surface of the carrier, and when used in excess, the cured product covers the zeolite particles to prevent the generation of off-flavor. Interfere with. When the suspension of the hydrophobic zeolite is sufficiently absorbed by the carrier by immersion or coating, and the excess hydrophobic zeolite suspension is removed and then dried, the hydrophobic zeolite particles enter the fine voids of the carrier and are fixed. .

The deodorizing catalyst is supported next to the carrier on which the hydrophobic zeolite is fixed, and the supporting treatment can be performed in the same manner as the case of supporting the hydrophobic zeolite. Since the hydrophobic zeolite is fixed first, the deodorizing catalyst hardly reaches the core of the carrier, and most of the catalyst is fixed to the surface layer of the carrier. If it is difficult to support the required amount of catalyst at one time, the catalyst impregnation and drying are repeated until an appropriate amount of catalyst is fixed.

In the deodorizing filter obtained by separately supporting the hydrophobic zeolite and the deodorizing catalyst as described above, the zeolite powder is supported at a higher density on the core than on the surface of the carrier, and the deodorizing catalyst is coated on the core of the carrier. It is carried at a higher density on the surface layer than on the part.

The deodorizing filter of the present invention can be used in exactly the same manner as a conventional deodorizing filter.
That is, it is used in combination with a suitable blowing means such as a fan so that the surrounding air flows in contact with the deodorizing filter.

[0020]

EXAMPLES Example 1 As a carrier, paper made of silica-alumina fiber (0.2 m thick) was used.

m, a porosity of 90%) and a honeycomb structure carrier / honeycle (product of Nichias Corporation) having a cell number of 350 / in ^{2} were prepared.

Hydrophobic zeolite ($SiO_2 / Al_2O_3 = 1$)

50 parts by weight, average particle size of 5 μm) are mixed with 100 parts by weight of silica sol (solid content: 30% by weight) and 80 parts by weight of water to form a slurry. (100 mm x 100 mm, length 15 mm in the ventilation direction), and then dried after removing excess slurry.

Next, the above-mentioned zeolite-supported carrier was added to a slurry comprising 100 parts by weight of Mn-Cu deodorizing catalyst / calolite (a product of Crows, USA), 80 parts by weight of silica sol (solid content: 30% by weight), and 280 parts by weight of water. Soak the

Dried. Thereafter, by immersing in the deodorizing catalyst slurry and drying again, the deodorizing catalyst carrying amount is 40 g / m^2 ,

A hydrophobic zeolite carrying amount of 10 g / m^2 and a residual porosity of 70

% Deodorizing filter was obtained.

This deodorizing filter was combined with a small fan for forced ventilation to constitute a deodorizer for treating indoor air in contact, and it was installed in a Western-style toilet and used continuously. A noticeable reduction in odor was observed as compared to before the use of the deodorizer, and the effect was maintained for 6 months or more. Further, even after use for 6 months, no off-flavor was felt from the deodorizing filter.

Comparative Example 1 A deodorizing filter was produced in the same manner as in Example 1 except that the carrier did not carry a hydrophobic zeolite, and was used as a deodorizer in the same manner as in Example 1. As a result, two months later, a specific odor different from the toilet odor was released from the deodorizing filter, although the original deodorizing performance was maintained.

[0025]

As described above, the deodorizing filter of the present invention does not generate secondary odor itself even when used for a long period of time, and can be used until the end of the life of the deodorizing catalyst. Thus, a reliable use effect can be obtained at a lower cost than conventional products.

続き Continued on the front page (58) Field surveyed

(Int.Cl. ^{6} , DB name) B01D 53/86-53/94 B01J 21/00-38/74 A61L 9/00

Cited By (4)

Publication number	Priority date	Publication date	Assignee	Title
Family To Family Citations				
WO1998033575A1 *	1997-01-31	1998-08-06	Takasago Thermal Engineering Co., Ltd.	Cleaning apparatus, filter and method of manufacturing the same
US20020177520A1 *	2001-03-30	2002-11-28	Nichias Co., Ltd.	Chemical filter
EP3124115B1 *	2014-03-27	2019-05-15	Hitachi Zosen Corporation	Honeycomb structure useful as exhaust gas cleaning catalyst, and its method for manufacturing
CN115551634A *	2020-05-15	2022-12-30	日挥通用株式会社	Deodorizing catalyst for refrigerator and deodorizing material for refrigerator using the same

* Cited by examiner, † Cited by third party, ‡ Family to family citation

Similar Documents

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US20030150820A1	2003-08-14	Filtering medium, method for making same
WO2001062307A1	2001-08-30	Apparatus for removing chemical substance
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JPH10165767A	1998-06-23	Odor preventing method for deodorizer
JP2000329365A	2000-11-30	Deodorizing element, deodorizing device, and refrigeration cycle provided with the same

Priority And Related Applications

Priority Applications (1)

Application	Priority date	Filing date	Title
JP7094268A	1995-03-29	1995-03-29	Deodorizing filter

Applications Claiming Priority (1)

Application	Filing date	Title
JP7094268A	1995-03-29	Deodorizing filter

Legal Events

Date	Code	Title	Description
2004-06-29	R250	Receipt of annual fees	**Free format text**: JAPANESE INTERMEDIATE CODE: R250
2005-05-10	R250	Receipt of annual fees	**Free format text**: JAPANESE INTERMEDIATE CODE: R250
2006-05-16	R250	Receipt of annual fees	**Free format text**: JAPANESE INTERMEDIATE CODE: R250
2007-04-10	R250	Receipt of annual fees	**Free format text**: JAPANESE INTERMEDIATE CODE: R250
2008-08-12	FPAY	Renewal fee payment (prs date is renewal date of database)	**Free format text**: PAYMENT UNTIL: 20090924 **Year of fee payment**: 10
2009-09-08	FPAY	Renewal fee payment (prs date is renewal date of database)	**Free format text**: PAYMENT UNTIL: 20100924 **Year of fee payment**: 11
2010-08-17	FPAY	Renewal fee payment (prs date is renewal date of database)	**Free format text**: PAYMENT UNTIL: 20110924 **Year of fee payment**: 12
2010-12-22	FPAY	Renewal fee payment (prs date is renewal date of database)	**Free format text**: PAYMENT UNTIL: 20110924 **Year of fee payment**: 12
2011-07-21	FPAY	Renewal fee payment (prs date is renewal date of database)	**Free format text**: PAYMENT UNTIL: 20110924 **Year of fee payment**: 12
2011-08-30	FPAY	Renewal fee payment (prs date is renewal date of database)	**Free format text**: PAYMENT UNTIL: 20120924 **Year of fee payment**: 13
2012-07-26	FPAY	Renewal fee payment (prs date is renewal date of database)	**Free format text**: PAYMENT UNTIL: 20120924 **Year of fee payment**: 13
2012-08-28	FPAY	Renewal fee payment (prs date is renewal date of database)	**Free format text**: PAYMENT UNTIL: 20130924 **Year of fee payment**: 14
2013-07-26	S531	Written request for registration of change of domicile	**Free format text**: JAPANESE INTERMEDIATE CODE: R313531

2013-08-05	R350	Written notification of registration of transfer	**Free format text**: JAPANESE INTERMEDIATE CODE: R350
2013-08-27	R250	Receipt of annual fees	**Free format text**: JAPANESE INTERMEDIATE CODE: R250
2014-09-24	LAPS	Cancellation because of no payment of annual fees	

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